

Important questions: 1

Class	CPI	Compiler A	Compiler B
A	1	5	4
B	2	2	3
C	3	1	2
Assuming a 2.5 GHz clock, calculate which compiler produces faster codes?			

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Compiler A's Total CPI = $(5 \times 1) + (2 \times 2) + (1 \times 3) = 12$ CPI

Compiler B's Total CPI = $(4 \times 1) + (3 \times 2) + (2 \times 3) = 16$ CPI

Compiler A's execution time = $12 \times 10^6 / 2.5 \times 10^9 = 0.0048 = 4.8$ ms (Faster)

Compiler B's execution time = $16 \times 10^6 / 2.5 \times 10^9 = 0.0064 = 6.4$ ms

Important questions: 2

- Consider the following two CPU designs:
CPU 1: Clock rate = 2.5 GHz, AVG. CPI = 1.6
CPU 2: Clock rate = 2.0 GHz, AVG. CPI = 1.2
- A program takes 2.2 billion instructions to execute.
 - 1) Compute the execution time on each CPU.
 - 2) Which CPU is faster and by what percentage?

Important questions: 2

- Consider the following two CPU designs:
 1. CPU 1: Clock rate = 2.5 GHz, AVG. CPI = 1.6
 2. CPU 2: Clock rate = 2.0 GHz, AVG. CPI = 1.2
- A program takes 2.2 billion instructions to execute.
 - 1) Compute the execution time on each CPU.
 - 2) Which CPU is faster and by what percentage?

1) Execution time of,

- CPU 1 = $(2.2 \times 10^9 \times 1.6) / 2.5 \times 10^9 = 1.40\text{s}$
- CPU 2 = $(2.2 \times 10^9 \times 1.2) / 2.0 \times 10^9 = 1.32\text{s}$ (Faster)

2) By what percentage?

- Slow / Fast = $1.40 / 1.32 = 1.060$ = CPU2 is 6.0% Faster $(1 - 1.060) * 100\%$

Important questions: 3

- Translate the following MIPS instructions into machine code:
 1. Addi t1, t2, 15
 2. Beq s1, s2, -8
 3. Lw t0, 32(sp)

Important questions: 3

1. A somewhat complete statement contains the five variables f,g,h,l and j:

$$f = (g + h) - (l + j)$$

What might an assembly compiler produce?

2. Show the IEEE 754 binary representation of the number -.075 in single and double precision.

3. Multiply 7×5 using all version of multiplication algorithm.

Consider the following C statements with five integer variables a , b , c , d , e :

$$\begin{aligned} a &= b + c; \\ d &= a - e; \end{aligned}$$

Translate the equations in MIPS assembly instructions. Calculate the Machine code. Assume, a , b , c , d , e will be $\$t0$, $\$t1$, $\$s0$, $\$s1$, $\$s2$.

Explain:

1. Suppose the variables a and e are stored in the same memory location by mistake. Explain what happens when these instructions execute.
 2. Write the expected values of a and d after execution in this scenario.
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The *first version of multiplication hardware* multiplies two integers x and y by performing repeated addition: adding x to an accumulator y times.

A student decides to reuse the accumulator memory for both x and z to save hardware registers. Draw the flowchart.

Explain:

1. Without computing exact numbers, explain why the result of $z = x * y$ may be incorrect in this scenario.
2. Describe a step-by-step sequence of additions showing how the accumulator changes if $x = 4$ and $y = 3$.
3. Suggest a creative fix that allows the hardware to still use minimal registers but always produces the correct result.